

Instalación y securización de una tienda online

Práctica

DESPLIEGUE DE APLICACIONES WEB

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La realización de esta práctica consiste en la instalación de una tienda online en una **instancia de AWS** (1GB Ram, 8GB HD y 1 vCPU) usando PrestaShop o Magento y securizando Apache2 con certificado autogenerado según vimos en la práctica anterior. Del mismo modo tienes libertad para usar otro gestor de contenidos para implementar la tienda online, justificando debidamente tu respuesta.

Los pasos que debes seguir son los siguientes: Instalación de una plataforma **LAMP** (GNU/Linux, Apache, MySQL y PHP).

- a. Configura el servidor para que disponga del entorno necesario para la posterior instalación de aplicaciones web. Debes generar un **certificado SSL** y asignarle el nombre miempresaX. Configura Apache para que implemente el protocolo **https**. Documenta cada paso.
- b. Habilita el sitio web (**http**) por defecto para que sea visualizado en el puerto 80.
- c. Configura el sitio web en el puerto 9000 para crear una redirección de **/aulavirtual** a la url <https://www.moodle.org>
- d. Realiza la instalación de una tienda virtual usando la aplicación **Prestashop** en tu servidor web.
- e. Realiza la instalación de una tienda virtual con **Prestashop** en un contenedor local **Docker** que debes crear en tu equipo local. Una vez terminada la instalación, despliega ese mismo contenedor en tu instancia de **AWS**.
- f. Establece los permisos necesarios en la instalación de Prestashop para que la aplicación sea segura.

Ayuda:

Para copiar el archivo de prestashop o magento si te lo descargas en tu equipo anfitrión:

```
scp -i "mi_certificado.pem" archivo_a_copiar.zip  
ubuntu@url_mi_servidor:/home/ubuntu
```

Para cambiar los permisos de ficheros y directorios:

```
find . -type f -exec chmod 644 -- {} +  
find . -type d -exec chmod 755 -- {} +
```

Cambia el usuario propietario de forma recursiva:

```
chown usuario-apache:usuario-apache -R /www/var/html
```

Os anoto los pasos necesarios para crear la base de datos directamente en mysql, sin necesidad de phpmyadmin:

1. `mysql -u root -p`
2. `CREATE DATABASE base_de_datos;`
3. `SHOW DATABASES;`
4. `GRANT USAGE ON mi_usuario.* TO base_de_datos@localhost IDENTIFIED BY 'contraseña_mi_usuario';`
5. `FLUSH PRIVILEGES;`

Instalación

Creamos la instancia:

y le asignamos las reglas de entrada:

Reglas de entrada (3)

🔄

Administrar etiquetas

🔍

Buscar

☐	Name	ID de la regla del gr...	Versión de IP	Tipo	Protocolo	Intervalo de puertos	Origen
☐	-	sgr-053927b7dca23dbd3	IPv4	HTTP	TCP	80	0.0.0.0/0
☐	-	sgr-0c97e4ddb2e7d6be6	IPv4	SSH	TCP	22	0.0.0.0/0
☐	-	sgr-094feca3fda29d0b7	IPv4	HTTPS	TCP	443	0.0.0.0/0

instalamos docker

creamos el docker con una Imagen LAMP

```
docker run -it --name prestashop-ssl -p 80:8080 -p 443:8443 antonio295/apache-php83-mariadb:v1
```

Añadimos nuestro usuario a la base de datos

```
MariaDB [(none)]> CREATE USER 'prestashop_manuel'@'localhost' IDENTIFIED BY 'manuel1234';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON prestashop_db.* TO 'prestashop_manuel'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> EXIT;
Bye
root@5b7e7672ab51:/etc/apache2/sites-enabled#
```

```
root@5b7e7672ab51:/etc/apache2/sites-enabled# service mariadb restart
* Stopping MariaDB database server mariadbd
* Starting MariaDB database server mariadbd
root@5b7e7672ab51:/etc/apache2/sites-enabled# mariadb
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 33
Server version: 10.11.13-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SELECT User, Host FROM mysql.user;
+-----+-----+
| User          | Host          |
+-----+-----+
| antonio       | localhost    |
| mariadb.sys   | localhost    |
| mysql         | localhost    |
| prestashop_manuel | localhost    |
| root          | localhost    |
+-----+-----+
5 rows in set (0.001 sec)

MariaDB [(none)]>
```

Cambiar los puertos por defecto de Apache

Editar el archivo de puertos

vim /etc/apache2/ports.conf

```
root@5b7e7672ab51: /etc/apache2

# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 8080

<IfModule ssl_module>
    Listen 8443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 8443
</IfModule>
```

Guarda el archivo.

vim /etc/apache2/sites-available/000-default.conf

```
root@5b7e7672ab51: /etc/apache2

<VirtualHost *:8080>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

Crear certificado:

Crea una clave privada RSA de 2048 bit.

```
openssl genrsa -out /etc/ssl/private/manuel.key 2048
```

c. Crea el certificado digital autofirmado usando la clave privada.

```
openssl req -new -key /etc/ssl/private/manuel.key -x509 -days 365 -out /etc/ssl/certs/manuel.pem
```

```
root@5b7e7672ab51: /etc/apache2

root@5b7e7672ab51:/etc/apache2# openssl genrsa -out /etc/ssl/private/manuel.key 2048
root@5b7e7672ab51:/etc/apache2# openssl req -new -key /etc/ssl/private/manuel.key -x509 -days 365 -out /etc/ssl/certs/manuel.pem
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:ES
State or Province Name (full name) [Some-State]:Sevilla
Locality Name (eg, city) []:Sevilla
Organization Name (eg, company) [Internet Widgits Pty Ltd]:manuel
Organizational Unit Name (eg, section) []:manuel
Common Name (e.g. server FQDN or YOUR name) []:manuel
Email Address []:manuel.avecilla.gavilan.al@iespoligonosur.org
root@5b7e7672ab51:/etc/apache2#
```

Activar el modulo SSL en Apache

```
a2enmod ssl
```

```
service apache2 restart
```

```
root@5b7e7672ab51:/etc/apache2# a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
    service apache2 restart
root@5b7e7672ab51:/etc/apache2#
```

Crear el VirtualHost HTTPS

```
vim /etc/apache2/sites-available/default-ssl.conf
```

```
<VirtualHost *:8443>
    ServerAdmin admin@localhost
    DocumentRoot /var/www/html

    SSLEngine on
    SSLCertificateFile /etc/apache2/ssl/manuel.crt
    SSLCertificateKeyFile /etc/apache2/ssl/manuel.key

    <Directory /var/www/html>
        AllowOverride All
    </Directory>

</VirtualHost>
```

```
<VirtualHost *:8443>
    ServerAdmin webmaster@localhost

    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf

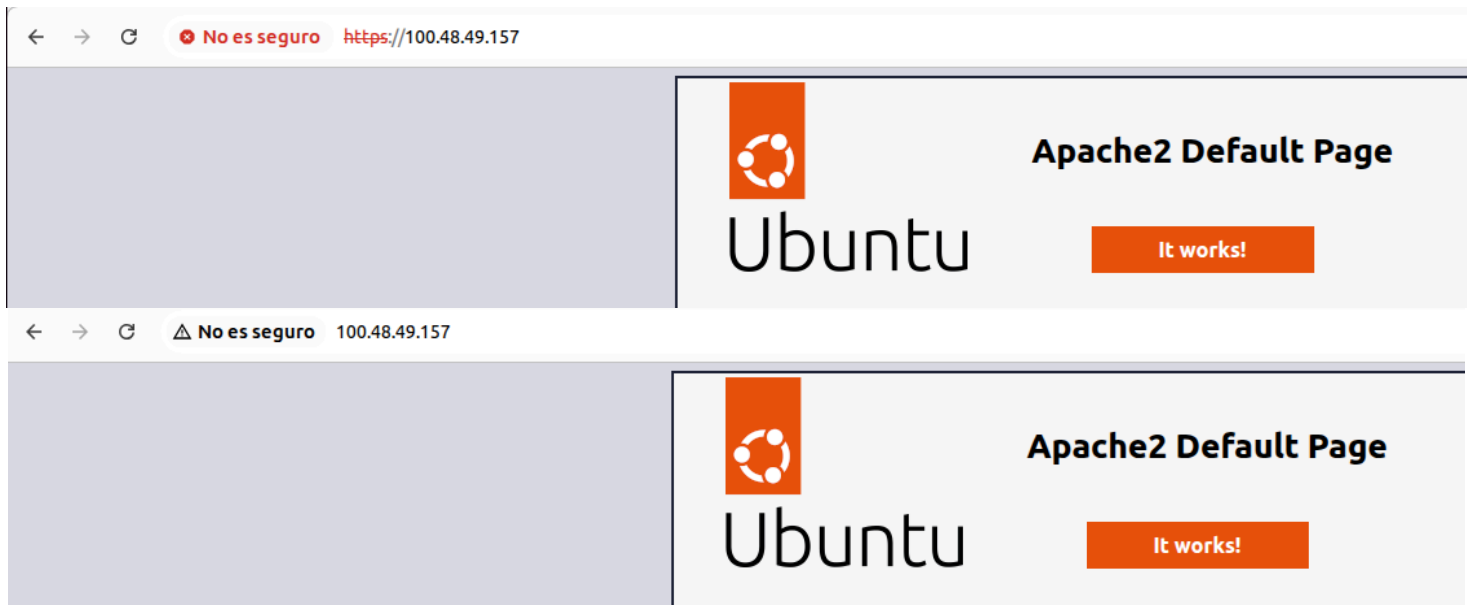
    # SSL Engine Switch:
    # Enable/Disable SSL for this virtual host.
    SSLEngine on

    # A self-signed (snakeoil) certificate can be created by installing
    # the ssl-cert package. See
    # /usr/share/doc/apache2/README.Debian.gz for more info.
    # If both key and certificate are stored in the same file, only the
    # SSLCertificateFile directive is needed.
    SSLCertificateFile      /etc/ssl/certs/manuel.pem
    SSLCertificateKeyFile   /etc/ssl/private/manuel.key
```

```
a2ensite default-ssl.conf
```

```
service apache2 restart
```

```
root@5b7e7672ab51: /etc/apache2
root@5b7e7672ab51:/etc/apache2# vim /etc/apache2/sites-available/default-ssl.conf
root@5b7e7672ab51:/etc/apache2# a2ensite default-ssl.conf
Enabling site default-ssl.
To activate the new configuration, you need to run:
  service apache2 reload
root@5b7e7672ab51:/etc/apache2# service apache2 restart
* Restarting Apache httpd web server apache2
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress th
is message
root@5b7e7672ab51:/etc/apache2# [ OK ]
```



PrestaShop

cd /var/www/html

wget https://assets.prestashop3.com/dst/edition/corporate/9.0.0-1.0/prestashop_edition_classic_version_9.0.0-1.0.zip

unzip prestashop_edition_classic_version_9.0.0-1.0.zip

rm prestashop_edition_classic_version_9.0.0-1.0.zip

rm index.html

chown -R www-data:www-data /var/www/html

chmod -R 755 /var/www/html

apt update

apt install -y php-intl

service apache2 restart

a2enmod rewrite

service apache2 restart

vim /etc/apache2/sites-available/000-default.conf

Escribe:

<Directory /var/www/html>

AllowOverride All

Require all granted

</Directory>

service apache2 restart